A non-isomorphic degree-three transport with a rational three-point fiber

F617 · full curve order 618 · target subgroup order 103 · cofactor 6 · eight same-source targets, 24/24 verified arm records

Source E: $y^2 = x^3 + 1$

ICV1:fp-617:0:618:0:unk:unk:r:b7eaec414c43

EC1P10Cfph8d1f0abcf065 · G = (66,264), order 103

ker φ(F617) = {O, (0,1), (0,616)}

Factor base: 12 distinct points after [6] projection

φ degree 3

Destination E′: $y^2 = x^3 + 590$

ICV1:fp-617:0:618:0:unk:unk:r:787b35c695fb

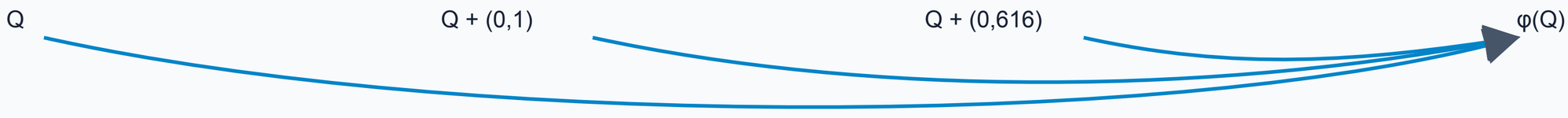
EC1P10Cfph9dc6fab612fb · G′ = (483,13), order 103

590 is not a sixth power in F617

Factor base: 12 distinct points after [6] projection

$\varphi(x,y) = ((x^3+4)/x^2,\ y(x^3-8)/x^3)$ for $x \neq 0$; the kernel maps to O

One exact target fiber, checked for each of eight targets



The map has image size 206 and every rational image point has exactly three preimages; on the order-103 subgroup, φ is injective.

Evidence: raw-v2.tsv, independent_v2.log, identities.tsv. Non-isomorphism uses the sixth-power criterion for short Weierstrass j=0 models.

The relation and rho bars count group additions and doublings only; this diagram does not assert a complete native-cost ECDLP advantage.